

PESU UNIVERSITY

BTECH AIML

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Software Engineering Lab 1 :

Section 1 :

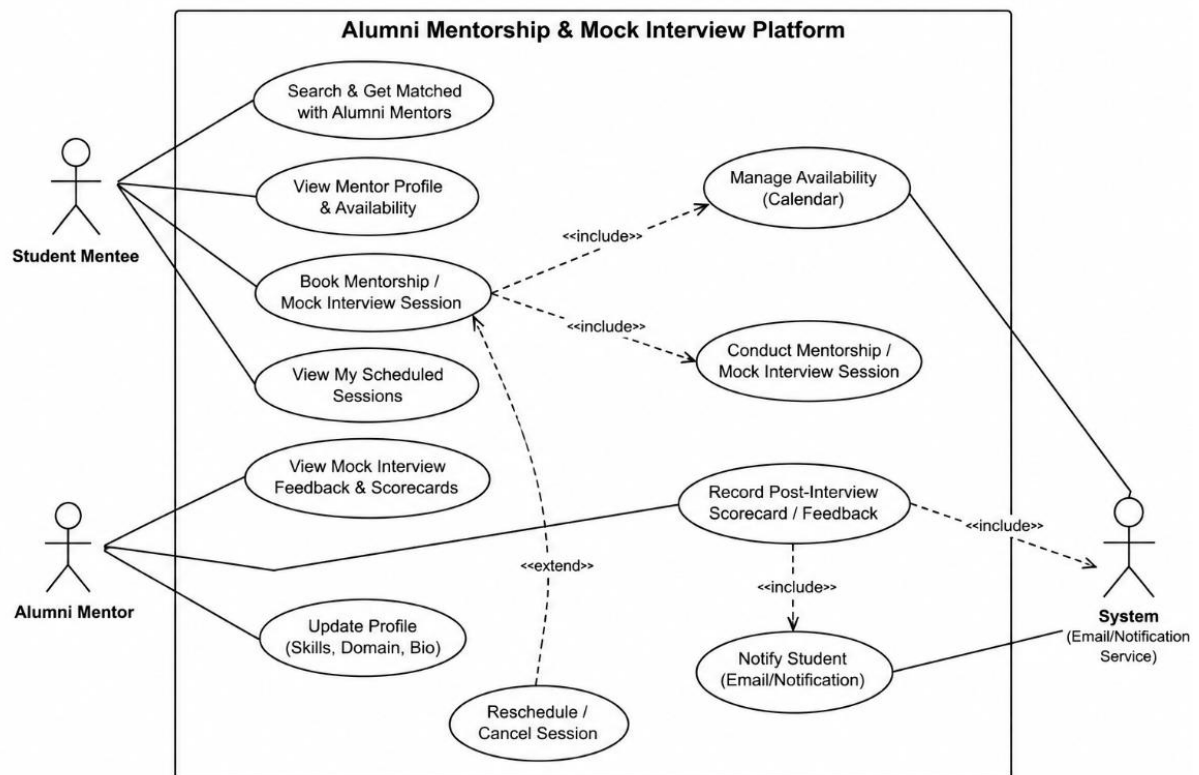
Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall: intelligently recommend alumni mentors to students based on the student's selected domain, skills, areas of interest, and mentorship requirements, and allow the student to select a suitable mentor for a session.	High	When a student enters valid domain and mentorship preferences, the system shall display a list of relevant alumni mentors. The results shall contain mentors whose listed domain matches the student's selected domain.	Ensures students are connected with alumni who have relevant industry knowledge and experience.
FR-002	Functional	The system shall: allow alumni mentors to create, view, and update their available dates and time slots for mentorship and mock interview sessions.	High	When a mentor adds or changes an available slot, the updated date and time shall be displayed in the mentor's availability calendar and shall be available for eligible students to book.	Allows mentors to control their availability and reduces scheduling conflicts.
FR-003	Functional	The system shall: allow a student to book a mentorship or mock interview session by selecting an available mentor and an available calendar slot.	High	When a student selects an available mentor and confirms an available time slot, the system shall create a booking containing the mentor, student, date, and time. The booked slot shall no longer be available for another student.	Provides a structured booking mechanism and prevents double-booking of mentorship sessions.
FR-004	Functional	The system shall: allow alumni mentors to record a structured scorecard containing ratings, comments, and feedback after completing a student's mock interview.	High	When a mentor submits a completed scorecard, the system shall save the ratings and feedback against the corresponding mock interview session. The submitted scorecard shall be accessible to the associated student.	Provides structured and actionable feedback to help students identify strengths and areas requiring improvement.
FR-005	Functional	The system shall: allow students to view their confirmed mentorship sessions, mock interview history, and completed post-interview scorecards.	Medium	When a student opens their mentorship dashboard, the system shall display their confirmed upcoming sessions and previously completed mock interviews. For completed interviews, the associated submitted scorecard and feedback shall be displayed.	Helps students track their mentorship activities and review feedback for future preparation.
NFR-001	Performance & Security	The system shall: protect student and alumni profiles, mentorship session information, and post-interview feedback according to applicable privacy guidelines while maintaining reliable performance under simulated peak load.	High	Security testing shall verify that authenticated users can access only the information they are authorized to view. Performance testing under simulated peak load shall show that core operations such as mentor search and session booking remain responsive.	Protects personal and feedback information while ensuring the platform remains dependable when many users access it simultaneously.
NFR-002	Usability & Availability	The system shall: provide a straightforward and responsive interface that allows students	High	Usability testing shall show that users can search for a mentor, view availability, and complete a	Makes the platform easy to use and reduces errors during mentor

		and alumni mentors to perform their primary tasks without unnecessary complexity.		session booking without external assistance. Calendar availability and booking status shall be clearly displayed.	selection and session scheduling.
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Section 2 :

UML Diagram



Section 3 :

Use Case Diagram :

Use Case ID: UC-03

Use Case Name: Book Mentorship / Mock Interview Session

Primary Actor: Student Mentee

Secondary Actor: Alumni Mentor

Preconditions:

1. The Student Mentee is authenticated and has an active profile.
2. The Alumni Mentor has a valid profile with their domain, skills, and mentorship details.
3. The system has identified the mentor as a suitable match for the student's selected domain and requirements.
4. The Alumni Mentor has at least one available date and time slot in their calendar.

Postconditions:

1. The mentorship or mock interview session is successfully booked.
2. The selected date and time slot is marked as unavailable for other students.
3. The booking details are stored in the student's and mentor's session records.
4. Both the Student Mentee and Alumni Mentor receive a booking confirmation.

Main Success Scenario:

1. Student Mentee logs into the Alumni Mentorship & Mock Interview Platform.
2. Student selects their required domain and mentorship/interview requirements.
3. System displays a list of recommended Alumni Mentors based on the student's requirements.
4. Student selects a suitable Alumni Mentor from the recommended list.
5. System displays the selected mentor's profile, expertise, and available calendar slots.
6. Student selects an available date and time for the mentorship or mock interview session.
7. System verifies that the selected slot is still available.
8. Student confirms the booking.
9. System creates the session with the Student Mentee, Alumni Mentor, date, time, and session type.
10. System marks the selected time slot as unavailable for further bookings.
11. System sends a confirmation notification to both the Student Mentee and Alumni Mentor.
12. Use case ends successfully.

Alternate Flow:

6a. Selected Time Slot Is No Longer Available

6a1. System detects that the selected time slot has already been booked by another student.

6a2. System informs the Student Mentee that the selected slot is no longer available.

6a3. System displays the Alumni Mentor's remaining available dates and time slots.

6a4. Student selects another available slot.

6a5. System verifies the newly selected slot and resumes at Step 8 of the Main Success Scenario.